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Qn1; Ranking Complexities from the Slowest to fastest.

- 1 - $O(n)$
- $O(\log n)$
- $O(n^5)$
- $O(n \log n)$
- $O(1)$

- Answer $\rightarrow O(1) < O(\log n) < O(n) < O(n \log n) < O(n^5)$

Qn 2; Finding Big-O complexity

a) $4n^2 + 20n + 9$

Answer; $O(n^2)$

b) $25n^3 + 15$

Answer: $O(n^3)$

c) $n^3 + 5 \log n + 6$

Answer; $O(n^3)$

d) $105 + 105n + 105n^2$

Answer: $O(n^2)$

e) $72n(n + 3n^2)$
 $= 72n^2 + 216n^3$

Answer: $O(n^3)$

f) $2^8 + 324n - 5$

Answer: $O(n)$

Qn 3: $O(n)$ Algorithm to sum 1 to n

$O(n)$ Algorithm to Sum 1 to n

function sumToN (n; integer): integer
begin

total := 0

for i := 1 to n do:

total := total + 1

end for

return total

end

Complexity: $O(n)$

Qn 4: Worst - Case Running Time

a) $X = 4$ is $O(1)$

The for loop runs number + 1 iterations, each doing $O(1)$ work.

Answer; Complexity = $O(\text{number})$

b) Worst case occurs when choice = 2 triggering the for loop
- The loop runs cycles + 1 iterations of $O(1)$ each

Answer; Complexity = $O(\text{cycles})$

c) $\text{increase}(\text{val}, \text{iter}) = O(\text{iter})$

$\text{Calculate}(\text{val}, \text{iter}) = O(\text{iter}^2)$

d) $\text{Calculate}(\text{val}, \text{iter})$; The outer loop runs $i = 1, 2, 3, \dots, \text{iter}$. Each iteration calls $\text{increase}(\text{val}, i)$, which costs $O(i)$

Total cost = $O(1) + O(2) + O(3) + \dots + O(\text{iter}) = O(1 + 2 + 3 + \dots + \text{iter})$
 $= O(\text{iter}(\text{iter} + 1)/2) = O(\text{iter}^2)$

Answer: Complexity = $O(n^2)$

Qn 5. Answers

i) $f(n) \leq c \cdot g(n)$

ii) $f(n) \geq c \cdot g(n)$

Qn 6. Filling the blanks

- We must find constants $c > 0$ and $k \geq 0$ such that for all $n \geq k$: $4n + 12 \leq cn$
The suitable constants; $c = 5$, $k = 12$

We need: $4n + 12 \leq 5n$, which simplifies to $12 \leq n$
This holds for all $n \geq 12$.

For all $n \geq 12$

Therefore

$4n + 12$ is $O(n)$

Qn 7:

We must find constants $c > 0$ and $k \geq 0$ such that for all $n \geq k$:

$$5n^2 - 16 \leq c \cdot n^2$$

Using the hint, let $c = 1$

$$5n^2 - 16 \leq n^2$$

$$4n^2 - 16 \leq 0$$

$$4n^2 \leq 16$$

$$n^2 \leq 4$$

$$n \leq 2$$

- This inequality is true only when $n \leq 2$. Since Big-O notation requires the inequality to hold for all sufficiently large value of n , $c = 1$ is not a suitable choice.

Now let $c=5$
substituting into the inequality

$$5n^2 - 16 \leq 5n^2$$

$$-16 \leq 0$$

This statement is always true for all values of n

Therefore, we can choose

$$c=5 \text{ and } k=1$$

Since there exist constants $c=5$ and $k=1$ such that

$$5n^2 - 16 \leq 5n^2$$

for all $n \geq 1$, we conclude that

$$5n^2 - 16 = O(n^2)$$

Answer: $c=5$, $k=1$, therefore $5n^2 - 16$ is $O(n^2)$